

CAN INTERACTION BECOME A PRIMARY SCIENTIFIC OBJECT?

A comparative object-of-science analysis across natural, behavioral, informational, and systems disciplines

Research paper

15 July 2026

Scope: interaction involving humans and computational systems, considered independently of particular interfaces or platforms

CENTRAL FINDING

Interaction can be treated as a primary scientific object because it denotes an observable, temporally extended pattern of reciprocal constraint and state change. Current evidence does not establish theoretical independence sufficient for an autonomous discipline. The justified next step is a dedicated comparative theoretical framework tested against integrated existing theories.

Abstract

This paper asks whether interaction itself can serve as a primary object of scientific explanation rather than merely a property of humans, machines, communication, cognition, or environments. It compares the scientific structure of twelve disciplines, reviews major ways interaction is conceptualized, and evaluates formal traditions that model interaction directly. The analysis distinguishes three claims that are often conflated: that interaction is real and measurable; that it can be selected as the focal explanandum; and that it requires an independent discipline. Relational and process-centered sciences such as ecology, linguistics, cybernetics, and systems theory show that a scientific object need not be a substance. Interaction meets the descriptive requirements of a scientific object when specified as a bounded trajectory of reciprocal causal influence among distinguishable systems: events can be observed, variables measured, mechanisms proposed, models falsified, and predictions tested. However, ecological psychology, dynamical systems, control theory, distributed cognition, activity theory, conversation analysis, HCI, computational interaction, process calculi, game theory, and reinforcement learning already supply much of the necessary ontology and formal machinery. No reviewed evidence demonstrates a general interaction-specific mechanism that is irreducible to a principled combination of those frameworks, and no cross-domain benchmark establishes superior predictions from an independent theory. The evidence therefore supports treating interaction as a primary scientific phenomenon and organizing a dedicated comparative theoretical program, but not yet an autonomous science. The paper ends with the minimum ontology and empirical tests needed to decide whether theoretical independence emerges.

Keywords: interaction; scientific object; HCI; cybernetics; systems theory; ecological psychology; computational interaction; ontology; process theory

1. Introduction

The expression “interaction” appears across nearly every science. Particles interact through fields; molecules interact through bonds and reactions; organisms interact with environments; speakers interact through language; people interact with machines; and processes interact through communication and synchronization. Ubiquity, however, does not establish scientific independence. A term may identify a mechanism inside another theory, a relation among already accepted entities, a descriptive label, or a genuine phenomenon whose regularities warrant direct explanation.

The central question is therefore not whether interaction matters. It is whether interaction can be made the subject of explanations in which the interaction trajectory, rather than either participant considered separately, is the principal dependent object. This requires a stricter test than demonstrating that interaction is observable. Scientific work routinely studies relations without granting each relation a separate discipline. Conversely, ecology, linguistics, and cybernetics demonstrate that a field may legitimately organize around relations, processes, or patterns rather than material substances.

This paper distinguishes a primary scientific object from an autonomous scientific domain. A primary object is a stable class of phenomena selected as the focal target of measurement and explanation. An autonomous domain additionally requires theoretical independence: characteristic mechanisms, explanatory questions, and cumulative models that cannot be replaced without loss by existing disciplines. The distinction permits a positive answer to the object question without presupposing a new discipline.

The paper's single contribution is a conditional determination: interaction is scientifically admissible as a primary object, but the evidence presently supports a distinct theoretical program rather than a distinct discipline. That determination is based on comparative structure, not on the existence of a community or the novelty of terminology.

2. What Constitutes a Scientific Object?

A scientific object is not simply anything that exists. It is an empirically tractable target stabilized by definitions, instruments, measurement conventions, models, and explanatory practices. Electrons, genes, species, grammatical structures, memory, and ecosystems differ ontologically, yet each becomes scientifically usable through operations that identify instances, distinguish variables, and connect observations to mechanisms.

Four conditions are necessary. First, an object must have identity conditions: researchers must be able to state what counts as an instance and where the object begins and ends. Second, it must support observation or warranted inference through an explicit observation model. Third, it must enter reproducible explanatory relations: changes in specified conditions should imply differences in measurements or distributions. Fourth, theories about it must be vulnerable to evidence. These conditions do not require perfect consensus or direct visibility. They require disciplined correspondence among definitions, observations, and claims.

A stronger condition applies when a proposed object is used to justify a new scientific domain. It must supply incremental explanatory power. Its characteristic variables or mechanisms must outperform, simplify, unify, or otherwise materially improve on established accounts. Institutional novelty, terminological fragmentation, and interdisciplinary inconvenience do not meet that standard.

For this study, interaction qualifies as a primary object only if it can be individuated independently of a named interface, measured as more than a list of participant properties, and modeled such that the trajectory has explanatory properties not captured by a static description of its participants. It qualifies as the basis of an independent domain only if these models require indispensable constructs or mechanisms absent from existing theories.

3. How Existing Sciences Define Their Objects

Table 1 compares scientific structure rather than disciplinary history. "Observer" identifies the investigator or measurement system; "observed" identifies the focal target. The rows are ideal

types: each discipline contains internal disagreement and subfields that cross the stated boundary.

Discipline	Primary scientific object	Observer / observed	Fundamental entities and relations	Typical explanatory questions	Primary mechanisms
Physics	Physical states and lawful changes of matter, energy, fields, space, and time.	Observer: instrumented investigator or detector. Observed: physical systems and events.	Particles, fields, spacetime, energy, momentum; forces, symmetries, conservation, interaction.	What changes? Under what invariances and laws? What produces motion, structure, or transition?	Dynamical laws, forces/fields, quantum interactions, conservation and symmetry breaking.
Chemistry	Composition, structure, properties, and transformations of matter.	Observer: chemist and analytical apparatus. Observed: substances, molecular systems, reactions.	Atoms, molecules, ions, bonds, phases; reaction and equilibrium relations.	Why does a substance have these properties? Which transformations occur, at what rates and yields?	Bonding, electron structure, thermodynamics, kinetics, catalysis, molecular collision.
Biology	Living systems and the processes that sustain, reproduce, develop, and diversify them.	Observer: biological investigator. Observed: molecules, cells, organisms, lineages, populations.	Cells, organisms, genes, traits, populations; inheritance, regulation, competition, cooperation.	How is life organized, maintained, developed, inherited, and changed?	Evolution, gene expression, metabolism, regulation, development, selection and drift.
Ecology	Relations among organisms and between organisms and environments across levels.	Observer: ecologist with field, experimental, remote, or model-based instruments. Observed: populations, communities, ecosystems and flows.	Organisms, populations, resources, habitats; predation, competition, mutualism, energy and matter flow.	What determines distribution, abundance, persistence, and ecosystem organization?	Selection, resource limitation, trophic interaction, disturbance, feedback, dispersal, succession.
Linguistics	Language as structured human capacity, system, and use.	Observer: linguist, corpus, experiment, field elicitation. Observed: utterances, judgments, practices, structures.	Phonological, morphological, syntactic, semantic and pragmatic units; combinatorial and contextual relations.	What structures are possible? How are form, meaning, use, acquisition, and change related?	Grammar, compositionality, inference, learning, interactional organization, historical change.
Psychology	Mind and behavior of humans and other animals.	Observer: researcher and behavioral/physiological instruments. Observed: behavior and inferred mental processes.	Percepts, memories, beliefs, goals, actions; causal and statistical relations among organism and context.	Why do individuals perceive, learn, decide, feel, and act as they do?	Perception, attention, memory, learning, motivation, decision, development, neural and social processes.

Discipline	Primary scientific object	Observer / observed	Fundamental entities and relations	Typical explanatory questions	Primary mechanisms
Human Factors	Human capabilities, limitations, and performance within work and technological systems.	Observer: analyst, experimenter, or operational monitoring system. Observed: person-system performance.	Humans, tasks, tools, organizations; workload, control, error, coordination and constraint.	How do system conditions shape safety, effectiveness, reliability, and well-being?	Attention and workload, control loops, skill-rule-knowledge behavior, automation, joint cognitive work.
Human-Computer Interaction	Use, design, experience, and consequences of interactive computing.	Observer: HCI researcher, designer, logs, laboratory or field methods. Observed: people, artifacts, practices and interaction episodes.	Users, computational artifacts, tasks, representations, actions, contexts; usability and experiential relations.	How do people use and experience computing, and how can interactive systems be explained or designed?	Cognitive, ecological, social, organizational, economic and computational mechanisms; design interventions.
Cognitive Science	Cognition and intelligence across biological and artificial systems.	Observer: interdisciplinary investigator and behavioral, neural, linguistic or computational instruments. Observed: cognitive systems and performances.	Representations or dynamical states, algorithms, bodies, environments, agents; computation, inference and coupling.	How is intelligent behavior generated, learned, represented, embodied, and distributed?	Computation, representation, neural processing, Bayesian inference, dynamical coupling, embodied action.
Information Science	Properties, behavior, flow, organization, retrieval, use, and effects of information.	Observer: information scientist and documentary/computational systems. Observed: information objects, users, institutions and processes.	Documents, signals, records, metadata, users, collections; relevance, retrieval, communication and transformation.	How is information generated, organized, communicated, found, interpreted, and used?	Indexing, classification, retrieval, communication, information behavior, social and technical mediation.
Cybernetics	Control and communication in animals, machines, and organizations.	Observer may be first-order external analyst or second-order participant-observer. Observed: regulatory systems and feedback relations.	Systems, signals, goals, disturbances, sensors, controllers; feedback, requisite variety, circular causation.	How does a system regulate itself, maintain variables, adapt, or communicate under disturbance?	Negative/positive feedback, control, estimation, communication, adaptation and recursion.
Systems Theory	Organization, relations, boundaries, emergence, and change in systems across domains.	Observer: modeler who declares system boundary and level. Observed: organized wholes and component relations.	Components, boundaries, states, inputs/outputs, networks; hierarchy, coupling, emergence and constraint.	How do organization and relations generate system-level behavior across scales?	Feedback, network structure, hierarchy, self-organization, nonlinearity, emergence and adaptation.

The comparison yields three conclusions. First, mature sciences are not distinguished solely by ontological type. Physics and chemistry emphasize substances and states; ecology emphasizes relations and populations; linguistics emphasizes structured capacities and products; cybernetics emphasizes circular processes. A relational phenomenon is therefore not disqualified from primary status.

Second, interaction is already a primary term inside several disciplines. Ecology directly studies organism-environment and organism-organism relations. Human factors officially defines itself around interactions among humans and system elements. HCI explicitly studies the human-computer interaction process. Cybernetics treats communication and feedback loops as the central phenomenon. Interaction is therefore not scientifically homeless.

Third, disciplines differ less by exclusive ownership of entities than by characteristic explanatory questions, boundary conventions, mechanisms, and evidence. Biology uses physical and chemical entities but asks biological questions. Ecology uses organisms already studied by biology but makes their relations and population-level consequences focal. The relevant test for interaction is consequently whether making the trajectory focal produces distinctive, cumulative explanations that are not merely renamed ecology, cognition, control, communication, or system dynamics.

4. How Interaction Is Currently Treated

4.1 Competing ontological roles

- **Object:** The interaction episode or trajectory is itself bounded, compared, and explained. Common in conversation analysis, interaction analysis, HCI studies, and some computational models.
- **Process:** Interaction is reciprocal influence unfolding over time. This is dominant in dynamical systems, ecological psychology, enactivism, cybernetics, and situated action.
- **Relationship:** Interaction denotes a relation between entities, often without asserting a separate object. Ecology, sociology, and network analysis frequently use this form.
- **Mechanism:** Interaction is a causal means through which outcomes arise: molecular collision, feedback, social influence, signaling, or human-machine coordination.
- **Property:** A system is described as interactive, coupled, responsive, or open. The property belongs to a system architecture rather than to an episode.
- **Outcome:** Interaction is measured through engagement, coordination, influence, or exchange produced by prior conditions.
- **Metaphor:** “Dialogue,” “conversation,” “partnership,” and “ecology” are sometimes transferred to systems without the commitments of the source theory. Metaphorical use is productive for design but insufficient for scientific identity.

4.2 Philosophical assumptions

Representational accounts treat interaction as information exchange among systems that possess internal states, goals, and mappings. Classical cognitive HCI, information-processing psychology, agent models, and many semantic ontologies take this form. Their strength is decomposition and

prediction; their risk is treating context and material organization as inputs to an otherwise self-contained processor.

Ecological and enactive accounts treat interaction as organism-environment coupling in which perception and action are co-defined. Affordances are relational possibilities for action rather than properties of an object alone. Their strength is direct treatment of embodiment and environment; their risk is underspecifying symbolic interpretation, institutional structure, or computational architecture.

Dynamical and cybernetic accounts treat interaction as coupled state change, feedback, regulation, and mutual constraint. They provide precise temporal and mathematical tools. Their risk is explanatory thinness when goals, meanings, norms, and historical practices are compressed into state variables without an account of their constitution.

Situated, distributed, and activity-theoretical accounts place interaction within material, social, historical, and organizational systems. They explain how action is mediated by artifacts, rules, communities, plans, representations, and division of labor. Their strength is boundary expansion; their risk is difficulty producing compact cross-domain predictions.

Pragmatic and interactional linguistic accounts treat meaning as publicly achieved through sequences of action, inference, turn-taking, repair, and common ground. They show that an utterance's meaning is not exhausted by its encoded content. Their risk, for a general interaction science, is domain restriction to communicative and socially accountable action.

Formal computer science treats interaction as communication, synchronization, observation, and response among computational processes or between system and environment. Process calculi, automata, game semantics, protocol models, and coalgebra provide representation-independent descriptions with strong compositionality and verification. They generally abstract away from human perception, learning, embodiment, and culture.

5. Systematic Literature Review

5.1 Review design

The review was structured as a theory-oriented scoping review rather than a meta-analysis. Searches targeted the terms interaction theory, interaction ontology, formal interaction model, interaction grammar, interaction primitives, interaction semantics, interaction machines, computational interaction, agent-environment coupling, social interaction, conversation analysis, process algebra, game semantics, cybernetics, ecological psychology, distributed cognition, situated action, activity theory, and human-computer interaction. Sources were selected when they proposed definitions, ontologies, formal mechanisms, or explicit boundaries, rather than merely reporting a designed system.

Priority was given to primary or field-defining sources: Wiener on cybernetics; von Bertalanffy on general systems; Gibson on affordances and ecological information; Card, Moran, and Newell on quantitative HCI; Suchman on situated action; Hutchins on distributed cognition; Sacks, Schegloff, and Jefferson on turn-taking; Milner and Hoare on process calculi; Abramsky and

colleagues on game semantics; and contemporary work on computational interaction and computational rationality. Professional definitions from the International Ergonomics Association, ACM SIGCHI, American Chemical Society, Ecological Society of America, Linguistic Society of America, and American Psychological Association were used only to characterize declared disciplinary objects.

The review sought contradictory evidence in two directions. Evidence against primary-object status included accounts that reduce interaction without loss to participant states, communication, cognition, control, or task performance. Evidence against a new discipline included frameworks that already supply a complete-enough ontology, mechanisms, measurement, and prediction. Evidence supporting primary-object status included relational effects, path dependence, emergence, and coordination patterns that cannot be inferred from participant properties considered independently.

This is not an exhaustive database review. Search coverage was broad across traditions but uneven across languages and publication venues. The assessment therefore concerns the theoretical record located and the standards it meets, not the absence of every possible prior formulation.

5.2 Major formalization traditions

Cybernetics and control theory. Interaction is a closed causal loop involving plant, controller, sensor, goal/reference, disturbance, and feedback. Strong on measurable dynamics, stability, adaptation, and intervention. Weak on semantic interpretation and social constitution unless supplemented.

Dynamical systems and coordination dynamics. Interaction is coupled evolution in a joint state space. Strong on emergence, synchronization, phase transitions, and multi-scale time. Weak when state variables lack independently validated psychological or social meaning.

Ecological psychology and enactivism. Interaction is organism-environment reciprocity; action possibilities are relational and perception is oriented toward action. Strong on embodiment, environmental structure, and mutuality. Weak on formal semantics, institutions, and many computational representations.

Conversation analysis and interactional linguistics. Interaction is sequentially organized social action, including turn construction, adjacency, preference, repair, and accountability. Strong on endogenous evidence and public organization. Limited generality beyond socially interpretable conduct.

Activity theory, situated action, distributed cognition. Interaction is mediated action within historically developed activity systems and distributed cognitive arrangements. Strong on artifacts, context, organizational roles, and development. Formal predictive comparison remains difficult.

Classical cognitive HCI. Interaction is decomposed into goals, operators, methods, selections, perceptual/motor/cognitive processors, and task structures. Strong on quantitative prediction for practiced tasks. Limited for open-ended interpretation and social change.

Computational interaction and computational rationality. Interaction is modeled using optimization, probabilistic inference, control, machine learning, cognitive models, and resource-bounded rationality. Strong on explicit generative models and design-relevant prediction. Often requires task-specific objective functions and representations.

Process calculi, automata, protocol theory, and interaction machines. Interaction is communication and state transition among open processes. Strong on compositional semantics, equivalence, liveness, safety, and verification. Human meaning and learning are normally outside scope.

Game theory and game semantics. Interaction is strategic or logical exchange among players/system and environment. Strong on incentives, information structure, strategies, and compositional meaning. Assumptions about rationality or formal moves can be restrictive.

Multi-agent and semantic ontologies. Interaction is represented through agents, actions, commitments, roles, messages, services, and protocols. Strong on machine-readable interoperability. Ontologies usually encode domain commitments rather than establish natural scientific mechanisms.

6. Comparative Analysis

6.1 Does interaction have identity conditions?

Interaction is not individuated merely by temporal co-occurrence. Two systems may change simultaneously without interacting. A defensible minimum criterion is counterfactual reciprocal dependence: during a bounded interval, a state or action of at least one participant alters the probability distribution over subsequent states, observations, or actions of another, and the resulting change can in turn condition later behavior in the joint trajectory. Reciprocity may be delayed, asymmetric, intermittent, mediated, or distributed; it need not occur in every adjacent event.

This definition separates interaction from one-way influence while allowing unequal agency. It also makes boundary selection explicit. An episode is bounded by an analytic rule such as task onset and completion, stabilization, conversational sequence closure, loss of causal coupling, or an experimentally imposed interval. Different boundaries may be legitimate for different questions, as in ecology and systems science, but they must be declared and tested for robustness.

The interaction object is therefore a trajectory over a joint state space, together with the coupling relations that generate it. It is not a third substance inserted between participants. Its identity resembles a reaction, conversation, control loop, or ecosystem process: a structured occurrence whose properties depend on relations and history.

6.2 Can interaction-level properties be non-reducible?

Several established phenomena support interaction-level description. Coordination can exhibit synchrony, entrainment, turn organization, escalation, deadlock, mutual adaptation, repair, and convention formation. These properties are not attributes of either participant alone. They

depend on coupling topology, timing, information availability, and history. Knowing each participant's isolated capability is generally insufficient to determine the joint trajectory.

Non-separability does not prove ontological independence. The joint behavior may still be explained completely by participant models plus coupling laws. In physics, a collision is a legitimate object of analysis even when the theory reduces it to fields and particles. Likewise, interaction can be primary for measurement and explanation without requiring fundamental primitives beyond agents, environments, states, actions, and coupling.

The strongest case for theoretical independence would be a law or mechanism whose explanatory role cannot be preserved by any reasonable decomposition into participant dynamics and established coupling theories. The reviewed literature does not yet provide such a result. Emergence, path dependence, and circular causation motivate interaction-level models, but systems theory, cybernetics, dynamical systems, ecology, and social interaction research already accommodate them.

6.3 Criteria assessment

Criterion	Evidence supporting	Evidence limiting	Assessment
Identifiable natural phenomena	Reciprocal causal episodes occur across biological, social, and computational settings; coupling can be operationally bounded.	Boundaries and granularity are observer-relative; “interaction” is too broad without a domain and identity rule.	Satisfied conditionally.
Observable events	Actions, messages, state changes, timing, feedback, repair, and outcomes can be recorded.	Meaning, belief, intention, and perceived opportunity are latent and require observation models.	Satisfied.
Measurable variables	Latency, transition probabilities, information flow, synchrony, error, utility, workload, learning, and coordination are measurable.	Measures are tradition-specific and often not invariant across contexts.	Satisfied, not yet standardized.
Reproducible behavior	Regularities exist in motor performance, turn-taking, feedback control, learning, and protocol behavior.	Open social settings and adaptive systems can be historically contingent and nonstationary.	Partly satisfied.
Explanatory mechanisms	Feedback, affordance, inference, learning, coordination, communication, constraint, and strategic adaptation are established.	Mechanisms come from adjacent disciplines; no uniquely interaction-specific mechanism is demonstrated.	Satisfied through integration.
Predictive capability	GOMS/KLM, control models, process models, computational rationality, game theory, and dynamical models predict bounded classes.	No single theory predicts heterogeneous interaction across tasks, modalities, cultures, and time scales.	Partly satisfied.
Empirical testability	Models imply intervention, trajectory, timing, error, transfer, and adaptation predictions.	Broad verbal interaction frameworks can evade decisive comparison if boundary and measures are unspecified.	Satisfied for formalized claims.
Theoretical independence	Interaction-level variables	Existing frameworks already	Not established.

Criterion	Evidence supporting	Evidence limiting	Assessment
	organize phenomena missed by isolated participant descriptions.	represent coupling, communication, coordination, emergence, and learning. Incremental superiority is unshown.	

7. Arguments Supporting Interaction as a Scientific Object

Relational objects are scientifically legitimate. Ecology treats relationships among organisms and environments as its declared object; cybernetics treats feedback and communication as central; linguistics studies structured relations between form, meaning, and use. Scientific objects need not be material substances.

The trajectory contains information absent from snapshots. Sequence, latency, mutual adaptation, repair, and path dependence cannot be reconstructed from static participant attributes alone. The ordering of events can change outcomes even when initial and final states match.

Interaction supports intervention. Researchers can manipulate coupling strength, feedback delay, action availability, information asymmetry, representation, incentives, and protocol rules, then measure predicted changes in trajectories. This supports causal rather than merely descriptive inquiry.

Interaction admits multiple formal representations. State-transition systems, differential equations, stochastic processes, process algebras, games, information-theoretic measures, and generative cognitive models can represent interaction without dependence on a particular interface.

Cross-domain regularities are plausible. Feedback, coordination, exploration, adaptation, breakdown, repair, and convention formation recur in biological, social, and computational systems. Whether these regularities support a unified theory is an empirical question, but they define a coherent comparative program.

Observer participation can be modeled. Second-order cybernetics, ethnomethodology, and measurement theory show that observer dependence need not invalidate an object. It requires explicit boundary, instrument, coding, and reflexivity assumptions.

8. Arguments Against

Excessive extension. If interaction includes every causal relation, the object becomes coextensive with science and loses discriminatory value. A domain must restrict participants, coupling, scale, and explanatory questions.

Redescription without mechanism. Calling a sequence an interaction does not explain it. Established mechanisms - perception, learning, control, communication, incentives, norms, and computation - may already provide all causal content.

Observer-imposed boundaries. Episodes, agents, and environments may be analytically selected rather than naturally segmented. Different choices can produce different interaction descriptions and apparent laws.

Heterogeneous mechanisms. Molecular interaction, social negotiation, and human-computer use may share only an abstract pattern of mutual influence. A theory general enough to include all may become too weak to predict any.

Existing integrative frameworks. Systems theory, cybernetics, ecology, cognitive science, distributed cognition, and HCI already study organized relations across components. A new framework must beat their principled combination, not isolated caricatures.

No demonstrated predictive increment. The literature contains strong local models but no accepted benchmark in which a general interaction theory yields replicated out-of-domain predictions superior to integrated existing models.

Institutional criteria are not scientific criteria. A journal, program, ontology, or professional identity can be useful without establishing a new natural kind or autonomous explanatory domain.

9. Attempted Falsification: Are Existing Theories Already Complete?

The strongest negative hypothesis is that an interaction episode is fully explained by: (1) participant state-transition models; (2) an environment model; (3) communication and observation channels; (4) coupling rules; and (5) established mechanisms for learning, control, decision, and social organization. On this view, “interaction” is a convenient projection of an already complete joint model.

Formal computer science supports much of this hypothesis. Open systems can be modeled as automata or processes whose observable behavior is defined through communication with an environment. Process calculi provide composition, equivalence, safety, and liveness. Game semantics gives meaning through structured exchanges between system and context. Control and dynamical systems provide joint trajectories and feedback. Reinforcement learning and partially observable decision processes provide action, observation, reward, belief update, and adaptation. Multi-agent systems add strategic dependence and communication.

Behavioral and social theories fill additional gaps. Ecological psychology supplies relational action possibilities; cognitive science supplies perception, memory, inference, and learning; conversation analysis supplies sequence organization and repair; activity theory supplies motive, mediation, rules, community, and development; distributed cognition supplies representational coordination across people and artifacts; human factors supplies joint performance and safety; HCI provides domain methods and empirical phenomena.

Taken together, these frameworks can express every item in the candidate ontology. This is substantial evidence against an independent discipline. Yet expression is not equivalent to a completed explanation. The frameworks use incompatible boundaries, units, semantics, and validation practices. Their combination is usually verbal rather than generative, and it is rarely

tested as a composite baseline. The completeness claim therefore remains unproven in the same way as the independence claim.

A decisive falsification program would construct the strongest integrated baseline available, not compare a new theory against one narrow field. If participant-plus-coupling models predict interaction trajectories, transfer, breakdown, and intervention outcomes across domains as well as any interaction-centered theory with equal or lower complexity, the independent-framework hypothesis should be rejected. If systematic residuals remain and require a stable new mechanism, the positive case strengthens.

10. Minimum Ontology if Interaction Is the Focal Object

The ontology below is a translation layer assembled from established vocabulary. It is not evidence for the conclusion and does not claim that every construct is fundamental. A new construct is warranted only where existing terms fail under empirical comparison.

Concept	Established equivalents and definition	Overlap / gap	New construct required?
Agent	Organism, actor, user, controller, process, participant: a system capable of state-dependent action or response.	Agency ranges from intentional human action to rule-governed process behavior; the intended level must be declared.	No.
Environment	Surround, context, world, task environment: conditions outside the focal agent boundary that affect or are affected by it.	Boundary is analytic and may move; social and computational elements can be agent or environment depending on question.	No.
State	System state, joint state, belief state, situation: variables sufficient for the model's future evolution at a chosen grain.	No state description is theory-neutral; latent human state requires measurement assumptions.	No.
Semantic state	Belief, interpretation, common ground, task meaning, content state: meaning-relevant distinctions attributed to an agent or public interaction.	The phrase risks reifying meaning as an internal object; distinguish public content, agent belief, and analyst coding.	No umbrella term needed.
Representation	External representation, sign, signal, display, symbol, inscription: a form preserving distinctions that can be perceived or manipulated.	Representation may be internal, external, conventional, indexical, or action-based; theory must state commitments.	No.
Opportunity	Affordance, action possibility, option, enabled action: a relation specifying actions possible for an agent under current constraints.	Physical feasibility, system permission, perceptual signaling, and agent belief are often conflated. These should be separated.	No new root construct; use typed action sets.
Action	Act, operation, move, control input, utterance: an event attributable to an agent that can alter joint state or information.	Intentional and nonintentional actions require different attribution rules.	No.
State transition	Reaction, event, update,	May be discrete, continuous,	No.

Concept	Established equivalents and definition	Overlap / gap	New construct required?
	transition: change from one state to another under actions, dynamics, and disturbances.	stochastic, or socially constituted.	
Observation	Percept, signal, measurement, feedback channel output: information made available to an agent or researcher.	Agent observation and researcher observation must be kept separate.	No.
Feedback	Action-contingent observation that can regulate subsequent behavior.	Not every observation is feedback; causal dependence on prior state/action matters.	No.
Adaptation	Learning, accommodation, policy update, plasticity, co-adaptation: persistent change induced by experience.	Change may occur in human, computational, environmental, or institutional components and at different time scales.	No.
Interaction episode	Conversation, trial, encounter, session, coupled trajectory: an analytically bounded interval of reciprocal causal influence.	Needed as the focal observational unit; boundaries remain claim-dependent.	A precise definition is required, but the vocabulary exists.
Evidence	Observation that discriminates among claims under an explicit observation and validity model.	Raw event logs are data, not self-interpreting evidence.	No.

11. Minimum Formal Object

At a declared time scale, let an interaction model contain agents $A = \{a_1 \dots a_n\}$, an environment E , a joint external state x_t , agent-specific observations $o_{i,t}$, actions $u_{i,t}$, internal or learned states $h_{i,t}$, and a transition rule $P(x_{t+1} \mid x_t, u_{1,t} \dots u_{n,t}, E_t)$. Observation functions map joint state to what each agent can detect. Policies map observations, internal state, goals, and constraints to action distributions. Learning rules update persistent agent state. Context variables specify social, institutional, linguistic, and physical conditions not otherwise included.

An interaction episode is a bounded trajectory $I = \langle x_0, o_0, u_0, x_1, \dots, x_T \rangle$ plus the declared coupling structure. It is interactive when cross-component dependence exceeds a specified null model: at least one participant's action or state changes the conditional distribution of another participant's later observations, states, or actions, and the resulting trajectory contains a return path or reciprocal dependence within the episode or across linked episodes.

This template is intentionally compatible with control systems, stochastic games, dynamical systems, process models, cognitive architectures, conversation sequences, and qualitative process tracing. It does not assert that every meaningful variable is numerical. It does require that each proposed variable have a stated observation or inference rule.

The model's scientific value lies in comparisons. Candidate interaction-level variables - coordination regime, repair state, convention, coupling strength, or path-dependent joint strategy

- must improve held-out prediction or causal explanation beyond participant and environment variables. Otherwise they are descriptive summaries rather than indispensable constructs.

12. Required Empirical Program

1. **Boundary robustness:** Test whether conclusions survive reasonable changes in episode boundary, agent definition, and temporal grain.
2. **Composite baseline comparison:** Compare interaction-centered models against the strongest integrated model from cognition, control, communication, learning, social organization, and system dynamics.
3. **Cross-representation invariance:** Hold task and semantic content constant while changing visual, auditory, linguistic, or physical representation. Determine which trajectory properties transfer.
4. **Cross-system prediction:** Fit models in one computational system and predict behavior in another without redefining core constructs.
5. **Longitudinal co-adaptation:** Measure human learning, computational adaptation, and environmental change jointly rather than treating one side as stationary.
6. **Intervention on coupling:** Manipulate feedback delay, observability, initiative, constraints, or communication topology and test predicted regime changes.
7. **Interaction-level residual test:** After modeling participant states and coupling, test whether stable residual structure remains that is captured by an interaction-specific construct.
8. **Measurement invariance:** Test whether constructs retain comparable meaning across cultures, languages, expertise levels, occupations, and access conditions.
9. **Failure criterion:** Reject the dedicated framework when established composite models achieve equal or better prediction, intervention selection, and explanatory compression with fewer assumptions.

13. Threats to Validity

Conceptual breadth is the principal threat. Interaction ranges from physical force exchange to social meaning-making. The paper mitigates this by restricting its operational conclusion to bounded reciprocal causal trajectories and by requiring domain declarations. The abstraction may still conceal mechanisms too heterogeneous for a common theory.

Source selection may favor traditions that already formalize interaction. Important work in anthropology, sociology, communication, robotics, organizational science, biology, and philosophy may not be fully represented. The conclusion is therefore calibrated to the evidence reviewed, not to an assertion of exhaustive coverage.

Disciplinary descriptions are simplified. Every listed field contains competing ontologies and methods. The comparison identifies dominant scientific structures rather than legislating a single correct definition.

Formal tractability can bias the analysis toward measurable state-transition accounts. Meaning, normativity, identity, power, and historical development may resist compact formalization while

remaining causally indispensable. The proposed empirical program therefore permits qualitative and mixed observation models, provided their inferential warrants are explicit.

Theoretical independence is difficult to prove negatively. Failure to find an irreducible mechanism does not show that none exists. Conversely, fragmentation and lack of integration do not prove that one does. The paper treats both claims as open to comparative testing.

The use of human-computational interaction as the motivating domain may overstate the generality of conclusions about interaction in other sciences. The proposed object definition is general, but the recommended research program should be evaluated first in domains where human and computational processes reciprocally adapt.

14. Discussion

The evidence resolves the central ambiguity by separating focality from fundamentality. Interaction can be focal because scientists can identify episodes, manipulate coupling, measure trajectories, and test mechanisms. It need not be fundamental in the metaphysical sense, and it need not displace agents, environments, cognition, communication, or computation. Reactions are primary objects for chemistry without being independent of physics; ecosystems are primary objects for ecology despite being composed of biological and physical entities. Scientific usefulness depends on explanatory organization, not ontological isolation.

Interaction becomes scientifically distinctive when the explanandum is the organization of mutual influence over time: why a trajectory stabilizes, diverges, coordinates, breaks down, repairs, transfers, or changes its own future coupling. These questions cannot be answered by listing interface events or participant attributes. They require joint state, information availability, action opportunity, feedback, history, and multi-scale adaptation.

Yet the ontology needed to ask those questions is not novel. Established theories already supply agents, environments, states, representations, affordances or action sets, actions, transitions, observations, feedback, adaptation, communication, strategy, context, and evidence. The scientific deficit is not missing nouns. It is the absence of a shared comparative framework in which different mechanisms make commensurable predictions about the same trajectories.

This finding argues against premature discipline formation. A new discipline would impose institutional boundaries before theoretical boundaries are demonstrated. It could encourage renaming established concepts, weaken engagement with mature methods, and make negative evidence harder to recognize. A dedicated theoretical framework is a smaller and more falsifiable commitment. It can remain inside HCI and cognitive science while drawing formal machinery from systems, control, ecology, linguistics, and computer science.

The framework should earn independence only through residual explanatory success. It must identify interaction-level variables that are stable across implementations, operationally distinct from participant variables, and predictively useful after strong baselines are included. The central benchmark is not whether a new vocabulary can describe more phenomena. It is whether it predicts or explains them with greater accuracy, transfer, or parsimony.

15. Conclusion

Interaction can be treated as a primary scientific phenomenon. It has identifiable instances when defined as a bounded trajectory of reciprocal causal influence; it produces observable events; it supports measurement, intervention, mechanism, prediction, and falsification; and mature sciences demonstrate that relations and processes can legitimately serve as primary objects.

The evidence does not support an autonomous science of interaction at present. Existing disciplines collectively provide the relevant entities, mechanisms, formalisms, and methods. Their integration is incomplete, but incompleteness of synthesis is not evidence of theoretical independence. No reviewed framework demonstrates a uniquely interaction-specific mechanism or replicated predictive increment over a strong composite baseline.

The justified scientific action is therefore to treat interaction as the focal explanandum of a dedicated comparative theoretical program. That program should use established terminology, define interaction episodes and observation models precisely, compare formal alternatives on shared data, and accept elimination if integrated existing theories prove sufficient.

FINAL ASSESSMENT

3. Interaction appears to satisfy the requirements of a distinct scientific object but additional theory is required.

References

- Abramsky, S. (2013). Semantics of interaction. arXiv:1312.0121.
- American Chemical Society. (2026). Chemistry is everywhere: What is chemistry?
- American Psychological Association. (n.d.). Psychology. APA Dictionary of Psychology.
- Ashby, W. R. (1956). An introduction to cybernetics. Chapman & Hall.
- Bertalanffy, L. von. (1968). General system theory: Foundations, development, applications. George Braziller.
- Borko, H. (1968). Information science: What is it? American Documentation, 19(1), 3-5.
<https://doi.org/10.1002/asi.5090190103>
- Card, S. K., Moran, T. P., & Newell, A. (1983). The psychology of human-computer interaction. Lawrence Erlbaum Associates.
- Chemero, A. (2009). Radical embodied cognitive science. MIT Press.
- Clark, H. H. (1996). Using language. Cambridge University Press.
- Dourish, P. (2001). Where the action is: The foundations of embodied interaction. MIT Press.
- Ecological Society of America. (n.d.). What is ecology?
- Engeström, Y. (1987). Learning by expanding: An activity-theoretical approach to developmental research. Orienta-Konsultit.
- Gibson, J. J. (1979). The ecological approach to visual perception. Houghton Mifflin.
- Goodwin, C. (2000). Action and embodiment within situated human interaction. Journal of Pragmatics, 32(10), 1489-1522.
- Hoare, C. A. R. (1985). Communicating sequential processes. Prentice Hall.

- Hollnagel, E., & Woods, D. D. (2005). Joint cognitive systems: Foundations of cognitive systems engineering. CRC Press.
- Hornbæk, K., & Oulasvirta, A. (2017). What is interaction? In Proceedings of CHI 2017 (pp. 5040-5052). ACM. <https://doi.org/10.1145/3025453.3025765>
- Hornbæk, K., Mottelson, A., Knibbe, J., & Vogel, D. (2019). What do we mean by “interaction”? An analysis of 35 years of CHI. *ACM Transactions on Computer-Human Interaction*, 26(4), Article 27. <https://doi.org/10.1145/3325285>
- Hutchins, E. (1995). *Cognition in the wild*. MIT Press.
- International Ergonomics Association. (2000). Definition and domains of ergonomics.
- Kaptelinin, V., & Nardi, B. A. (2006). *Acting with technology: Activity theory and interaction design*. MIT Press.
- Kelso, J. A. S. (1995). *Dynamic patterns: The self-organization of brain and behavior*. MIT Press.
- Linguistic Society of America. (n.d.). Who we are.
- Milner, R. (1989). *Communication and concurrency*. Prentice Hall.
- Newell, A., & Card, S. K. (1985). The prospects for psychological science in human-computer interaction. *Human-Computer Interaction*, 1(3), 209-242.
- Norman, D. A. (2013). *The design of everyday things* (Rev. ed.). Basic Books.
- Oulasvirta, A., Jokinen, J. P. P., & Howes, A. (2022). Computational rationality as a theory of interaction. *Proceedings of CHI 2022*. ACM. <https://doi.org/10.1145/3491102.3517739>
- Oulasvirta, A., Kristensson, P. O., Bi, X., & Howes, A. (Eds.). (2018). *Computational interaction*. Oxford University Press.
- Parasuraman, R., Sheridan, T. B., & Wickens, C. D. (2000). A model for types and levels of human interaction with automation. *IEEE Transactions on Systems, Man, and Cybernetics - Part A*, 30(3), 286-297.
- Reeves, S. (2015). Human-computer interaction as science. *Proceedings of the Fifth Decennial Aarhus Conference*, 73-84.
- Sacks, H., Schegloff, E. A., & Jefferson, G. (1974). A simplest systematics for the organization of turn-taking for conversation. *Language*, 50(4), 696-735.
- Simon, H. A. (1962). The architecture of complexity. *Proceedings of the American Philosophical Society*, 106(6), 467-482.
- Suchman, L. A. (1987). *Plans and situated actions: The problem of human-machine communication*. Cambridge University Press.
- Thelen, E., & Smith, L. B. (1994). *A dynamic systems approach to the development of cognition and action*. MIT Press.
- Varela, F. J., Thompson, E., & Rosch, E. (1991). *The embodied mind*. MIT Press.
- Wegner, P. (1997). Why interaction is more powerful than algorithms. *Communications of the ACM*, 40(5), 80-91.
- Wiener, N. (1948). *Cybernetics: Or control and communication in the animal and the machine*. Wiley.
- Winograd, T., & Flores, F. (1986). *Understanding computers and cognition*. Ablex.
- Zhang, J., & Norman, D. A. (1994). Representations in distributed cognitive tasks. *Cognitive Science*, 18(1), 87-122.